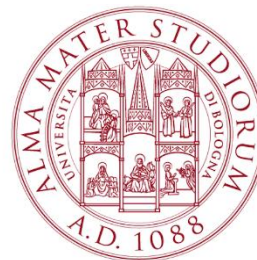


Building a Profile Hidden Markov Model for the Kunitz-type protease inhibitor domain

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Abstract

The Kunitz-type protease inhibitor domain plays a key role in numerous biological processes, particularly in the regulation of proteolytic activity. This study focuses on the characterization of the Kunitz domain and the development of predictive models for its detection using profile Hidden Markov Models (HMMs). The model was built from a manually refined structural alignment generated via PDBFold and implemented using the HMMER suite. Its performance was evaluated as a binary classifier on curated positive and negative datasets from UniProt/SwissProt, following a two-fold cross-validation strategy. Standard classification metrics were calculated to assess performance, including accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient (MCC).

At the optimal E-value threshold ($1e^{-3}$), the structure-based model achieved an MCC of 1.000 in Set 2, with zero false negatives, and an MCC of 0.9918 in Set 1. These results highlight the model's robust ability to detect Kunitz domains, particularly in borderline or weakly annotated sequences. Overall, our findings support the integration of structural information into HMM-based domain detection, providing a reliable strategy for accurate protein domain annotation.

Supplementary materials available at: https://github.com/sofianatale/LAB1_project

1. Introduction

1.1 Kunitz protease inhibitor domain

Kunitz-type domains (InterPro: IPR002223, Pfam: PF00014) are highly conserved protein motifs involved in inhibiting protease activity, primarily against serine proteases (Gene Ontology: GO:0004867), but can also target cysteine and aspartic proteases. These domains were first identified in the Bovine Pancreatic Trypsin Inhibitor (BPTI) and have since been found across a wide spectrum of organisms, ranging from microorganisms to mammals. Their potent inhibitory effects is crucial in diverse physiological processes including inflammation regulation, ion channel modulation, coagulation and tissue protection.

Structurally, the Kunitz domain consists of approximately 60 amino acid residues (~ 6 kDa), forming a compact $\alpha+\beta$ fold architecture. This conformation is maintained by the presence of six conserved cysteine residues that form three disulfide bridges (C1–C6, C2–C4, C3–C5), essential for

mediating interaction with target proteases through the formation of a protease-binding loop [1].

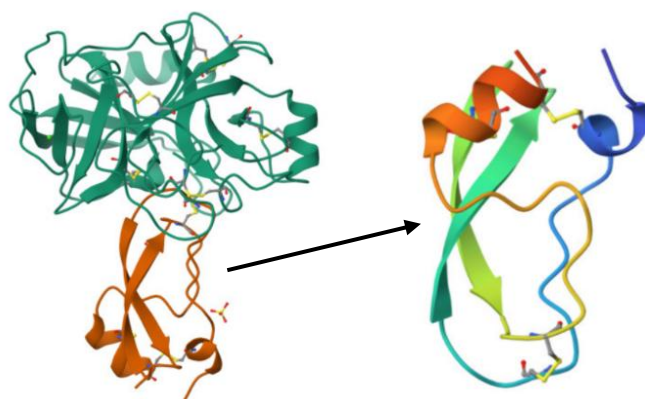


Figure 1: From PDB - Canonical structure of the Kunitz-type protease inhibitor domain. *a)* Crystal structure of the bovine pancreatic trypsin inhibitor (BPTI) in complex with rat anionic trypsin (PDB ID: 3TGI). The Kunitz domain (orange) is shown interacting with the protease (green). *b)* Isolated view of the Kunitz domain highlighting its conserved $\alpha+\beta$ fold stabilized by three disulfide bridges. This used as a reference for structural alignment in model construction.

Kunitz domain are found in various biologically relevant proteins, include the Tissue Factor Pathway Inhibitor (TFPI), where they contribute to anticoagulation; the Amyloid Precursor Protein (APP), linked to Alzheimer's disease; inter-alpha-trypsin inhibitors; types VI and VII collagen chains; as well as various toxins and protease inhibitors found in venoms and legume seeds [2]. Beyond their role in vertebrate, Kunitz domains have also evolve diverse function in invertebrates. In parasitic helminths, they protect against host digestive enzymes, while in hematophagous arthropods they act as anticoagulants. This wide functional spectrum across distant taxa underscores their biological relevance and the necessity of accurate computational methods to identify them [3].

However, challenges in detecting Kunitz domains persist due to high sequence variability, multiple domain copies, and inconsistent annotations in databases like UniProt. For this reason, Hidden Markov Models (HMMs) represent a valuable tool for their sensitive and specific identification.

1.2 Hidden Markov Models and structure-based domain detection

Hidden Markov Models (HMMs) are probabilistic models widely used in bioinformatics for analyzing biological sequences such as DNA, RNA, or proteins, by capturing both conserved and variable regions across evolution [4].

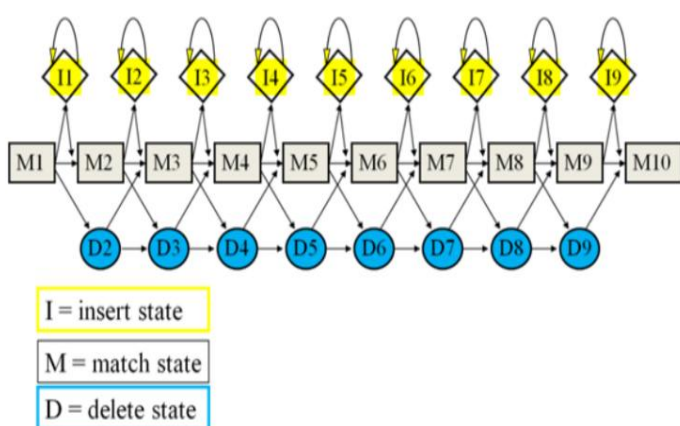


Figure 2: Adapted from *InterPro – Training Online*. Profile HMM built from a multiple sequence alignment. Each column of the alignment corresponds to a match state (M), while insertions (I) and deletions (D) are modelled explicitly to capture evolutionary variation. This probabilistic structure allows sensitive detection of conserved domains across divergent sequences.

In the context of protein domain detection, a specialized form known as profile HMM is trained on a multiple alignment of related sequences to encode the statistical properties of each position; specifically match (M), insert (I), and delete (D) states. This allows the model to explicitly account for evolutionary events such as insertions and deletions, enabling more sensitive and accurate recognition of homologous domains compared to traditional tools like BLAST [5].

While profile HMMs are typically trained on multiple sequence alignments (MSAs), this approach has limitations when dealing with highly divergent protein families. To overcome these, this project employs a structure-guided strategy, in which a multiple alignment is derived from the structural superposition of experimentally solved Kunitz-domain proteins. Structural alignment offers a more functionally informative representation by aligning residues based on their spatial positions rather than sequence similarity alone, preserving core secondary structure elements and conserved disulfide bridges essential to the Kunitz fold. The structural alignment was generated using PDBeFold [10] and manually refined to ensure consistency across key features.

This structure-based alignment forms the foundation of the profile HMM, ensuring that the model captures spatially conserved and functionally critical residues that may be missed by sequence-based approaches. Particularly in the "twilight zone" of sequence identity (20–30%), where homology is difficult to detect, structure-informed HMMs offer enhanced specificity, sensitivity, and biological interpretability. The resulting model computes log-odds scores for query sequences relative to a background model, providing statistically grounded predictions. Due to their capacity to model complex conservation patterns, structure-based HMMs are powerful tools for identifying remote homologs and poorly annotated instances of protein domains, as demonstrated in the case of the Kunitz-type protease inhibitor domain.

1.3 Aim of the project

The aim of this research is to develop a profile Hidden Markov Model (HMM) specifically tailored to detect the Kunitz protein domain. This endeavour involves several key stages: collecting relevant data from curated protein databases, constructing the model architecture, and ultimately applying it to annotate Kunitz domains within the SwissProt dataset. The goal is to create a robust model capable of identifying proteins that potentially contain Kunitz domains and evaluate the generalization capabilities of both models in identifying true Kunitz domains in a diverse set of protein sequences.

2. Materials and Methods

2.1 Sequence retrieval and Dataset assembly

To develop an accurate Hidden Markov Model (HMM) for the Kunitz-type protease inhibitor domain, protein sequences were collected from two major sources: UniProtKB/Swiss-Prot [6] and the Protein Data Bank (PDB) [7].

Sequences containing the Kunitz-type serine protease inhibitor domain (Pfam ID: PF00014) were identified using InterPro annotations [8] and downloaded in FASTA format from UniProtKB. A total of 395 reviewed sequences were retrieved, including 18 human proteins and 377 non-human entries proteins. These were saved separately (human_kunitz_sequences.fasta, non_human_kunitz_sequences.fasta) and merged into a single dataset (all_kunitz.fasta) to serve as the initial positive set.

For the negative dataset, the complete Swiss-Prot database (uniprot_sprot.fasta) was used. All entries matching the positive identifiers were removed to generate a pool of sequences assumed to lack the Kunitz domain. This background set was saved as sp_negs.fasta. The rationale behind this approach was to simulate a realistic classification scenario where the majority of proteins do not contain the domain of interest.

In parallel, 158 PDB entries annotated with PF00014 were retrieved via a custom export (rcsb_pdb_custom_report_20250505025420.csv) from the RCSB PDB. Filters were applied to retain only entries resolved by X-ray crystallography, with resolution ≤ 3.5 Å and chain lengths between 45 and

80 residues, reflecting typical Kunitz-domain structural characteristics.

2.2 Redundancy reduction and quality control

To reduce redundancy and ensure sequence diversity, the 158 PDB-derived sequences were clustered using CD-HIT at a 90% identity threshold (-c 0.9) [9]. This cutoff was chosen to eliminate nearly identical entries while preserving biologically meaningful diversity in the dataset. It prevents model overfitting and enhances generalization, which is critical when training probabilistic models such as HMMs. From this clustering, 25 non-redundant sequences were selected and stored in pdb_kunitz_rp.fasta.

To further ensure alignment quality, sequences were screened for outliers in terms of length and insertions. The goal was to prevent biased alignments caused by sequences with atypically long loops or extensions. One sequence, 2ODY:E, was found to be significantly longer than the rest and was excluded from the dataset. Additionally, the structural alignment revealed that 5JBT:Y, although similar in sequence, displayed poor structural compatibility with the rest of the dataset. It had a high RMSD value (2.92 Å) and an unusually short length (38 residues), suggesting it may represent a divergent or truncated domain. For these reasons, 5JBT:Y was also excluded from further analysis.

The curated set was then used to generate a list of PDB identifiers (tmp_pdb_efold_ids.txt), formatted for input into PDBeFold. This list served as the reference for the structure-based alignment.

However, structural data may still overlap with sequences from UniProt. To ensure that the model evaluation would be conducted on unseen data, a BLAST-based similarity search was performed. All sequences from all_kunitz.fasta showing $\geq 95\%$ identity and $\geq 50\%$ alignment coverage to any representative structural sequence were removed. This filtering eliminate near-duplicates, ensuring that the evaluation phase tested the model's ability to detect distantly related homologs. The remaining sequences were saved in ok_kunitz.fasta.

2.3 Model building

The 23 curated PDB sequences were structurally aligned using PDBeFold, which performs pairwise and multiple structural superpositions to identify spatially conserved residues based on 3D geometry. The initial alignment `pdb_kunitz_rp.ali` was manually inspected and refined to remove poorly aligned segments, adjust excessive gaps, and ensure consistent alignment length across all input chains.

The final curated alignment was saved as `pdb_kunitz_rp_formatted.ali` and served as input for the HMMER suite to construct a profile Hidden Markov Model. The resulting file, `structural_model.hmm`, encodes the statistical representation of the structural alignment, including match, insertion, and deletion states. The model includes 58 match states, corresponding to the conserved structural core of the Kunitz domain.

This structure-guided HMM was later used for large-scale sequence scanning and performance evaluation.

2.4 Cross-Validation strategy

To assess the performance and generalizability of the structural HMM, a two-fold cross-validation scheme was employed using both positive and negative datasets.

The filtered positive set (`ok_kunitz.fasta`)—containing sequences with no significant similarity to the structural training set—was randomly divided into two equally sized subsets: `pos_1.fasta` and `pos_2.fasta`.

Similarly, the background negative set (`sp_negs.fasta`) was randomly split into `neg_1.fasta` and `neg_2.fasta`, each containing the same number of entries as their corresponding positive subsets. Although the negative dataset is expected to contain few or no true Kunitz-domain sequences, balanced splitting ensured equal class sizes across validation folds.

Corresponding `.ids` files (for example `pos_1.ids`, `neg_1.ids`) were generated to track sequence identifiers and ensure reproducibility across evaluations. A custom script (`get_seq.py`) was used to automate the selection and formatting of sequence subsets. Care was taken to maintain separation between training and test data to avoid evaluation bias.

2.5 Domain scanning and prediction output

The trained HMM (`structural_model.hmm`) was applied to each test set using `hmmsearch`, a tool from the HMMER suite [11] designed to identify statistically significant matches between a profile HMM and a sequence dataset. The search was conducted with the `-max` and `-Z 1000` options to ensure consistent statistical calibration and enable accurate comparison of E-values across sequences.

For each sequence, the output included a full-sequence E-value, representing the probability of observing a match of equal or better score by chance. Lower E-values indicate stronger evidence of domain presence.

To ensure that every sequence was accounted for in the final evaluation, sequences absent from the `hmmsearch` output, typically due to extremely high E-value, were manually added with a placeholder value of 10.0. This step ensured that negative predictions were properly included in downstream classification. The results for each test subset were formatted into `.class` files containing:

- UniProt accession ID
- True class label (1 for Kunitz, 0 for non-Kunitz)
- Full-sequence E-value

Each fold produced two `.class` files (for example, `pos_1.class`, `neg_1.class`), which were subsequently merged into two comprehensive evaluation sets: `set_1.class` and `set_2.class`.

2.6 Performance evaluation metrics

Performance metrics were calculated using a custom Python script (`performance.py`) that parsed the `.class` files and computed key binary classification statistics across a range of E-value thresholds (from $1e-01$ to $1e-10$). The following metrics were calculated:

Accuracy (Q2): overall correctness of predictions

$$Q2 = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision (PPV): proportion of predicted positives that are true positives

$$PPV = \frac{TP}{TP + FP}$$

Recall (TPR): proportion of true positives correctly identified

$$TPR = \frac{TP}{TP + FN}$$

F1-score: harmonic mean of precision and recall

$$F1\ score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

Matthews Correlation Coefficient (MCC): balanced metric for binary classification, especially informative in imbalanced datasets

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

False Positive Rate (FPR): the proportion of actual negative sequences that were incorrectly predicted as positive by the model.

$$FPR = \frac{FP}{FP + TN}$$

False Negative Rate (FNR): the proportion of actual positive sequences that were missed by the model (i.e., classified as negative)

$$FNR = \frac{FN}{FN + TP}$$

The optimal E-value threshold was determined as the one maximizing MCC. This threshold was selected for final classification because MCC provides a robust, single-value summary that accounts for all four confusion matrix elements.

The performance metrics obtained, stored in `performance_set1_thresholds.txt` and `performance_set2_thresholds.txt`, demonstrate the high discriminative power of the structural HMM. In particular, high MCC values and the absence of false negatives at optimal thresholds underline the advantage of incorporating structural information into HMM construction.

3. Results and Discussion

3.1 Cross-validation performance

To assess the predictive capacity of the structural HMM trained on Kunitz-type domains, we employed a two-fold cross-validation approach designed to test the model on entirely unseen sequences in each round. This procedure ensured that the final evaluation reflected the model's ability to generalize beyond the training examples, particularly under the challenging conditions imposed by a markedly unbalanced dataset.

Fold 1	TP = 180, FN = 3 TN = 283,974, FP = 0
Fold 2	TP = 182, FN = 1 TN = 283,947, FP = 0

Table 1: Confusion matrix results per fold (E-value = $1e-5$).

Each validation fold consisted of 183 positive sequences and approximately 284,000 negatives. The model was evaluated separately on each set, ensuring a balanced testing strategy while maintaining a strong class imbalance reflecting a realistic biological context in which true domain-containing sequences are vastly outnumbered by non-hits. The model's predictions were evaluated using confusion matrices, allowing us to extract a set of performance metrics that included recall (sensitivity), precision, and the Matthews Correlation Coefficient (MCC), which is particularly appropriate for imbalanced classification tasks.

In Fold 1, the classifier identified 180 of the 183 positive sequences correctly, producing 3 false negatives and no false positives among the 283,974 negatives (TN = 283974, TP = 180, FN = 3, FP = 0). This resulted in a recall of 0.9836, precision of 1.000, and an MCC of 0.9918. The accuracy, measured as the

proportion of correctly classified sequences across the entire test set, was also optimal ($Q2 = 1.000$).

CONFUSION MATRIX – Set 1

True Label	Negative	True Negative 283974	False Positive 0
	Positive	False Negative 3	True Positive 180
		Negative	Positive
		Predicted Label	

Figure 4: Confusion matrix for Fold 1. The model was trained on subset B and evaluated on subset A at an E-value threshold of $1e^{-5}$, shows 180 true positives and 283,974 true negatives. Only 2 positive sequences were misclassified (false negatives), and no false positives were observed, indicating excellent specificity.

The performance in Fold 2 was even more favourable. Out of 183 positives, 182 were correctly predicted, with only a single false negative, again without any false positives. The confusion matrix (TN = 283947, TP = 182, FN = 1, FP = 0) translates to a recall of 0.9945, precision of 1.000, and an MCC of 0.9973, confirming a near-perfect performance even under strong class imbalance.

CONFUSION MATRIX – Set 2

True Label	Negative	True Negative 283947	False Positive 0
	Positive	False Negative 1	True Positive 182
		Negative	Positive
		Predicted Label	

Figure 5: Confusion matrix for Fold 2. The model was trained on subset A and tested on subset B using the same threshold ($1e^{-5}$). The classifier identified 181 true positives and 283,947 true negatives. However, 1 domain-containing sequence was misclassified as negative (FN = 1), and 0 non-Kunitz sequences were incorrectly classified as positives (FP = 0).

Such high MCC values, approaching 1 in both validation rounds, indicate that the model succeeds in distinguishing between the two classes in a highly consistent and balanced way. The complete absence of false positives across both folds is particularly remarkable and reinforces the strength of the structural features encoded in the profile. These results suggest that the model is not merely memorizing frequent patterns in positive sequences, but rather learning discriminative properties that are rare in unrelated proteins.

All classification metrics and confusion matrices in this section were computed using a more conservative threshold of $1e^{-5}$. This decision was made to reflect typical stringency levels applied in practical domain annotation pipelines and to emphasize the model's robustness under strict conditions, even though the optimal threshold yielding the highest MCC was identified at $1e^{-3}$ [Section 3.2].

Additionally, to explore how sensitive the model was to different E-value cutoffs, we ran comparative evaluations across a range of thresholds, including $1e^{-5}$ (a commonly used threshold in HMM applications) and $1e^{-9}$ (a highly stringent value). The optimal threshold yielding the highest MCC was $1e^{-3}$, balancing sensitivity and specificity effectively. Although minor variations were observed, the results remained robust and stable across thresholds, with negligible differences in key metrics. This consistency suggests that the HMM captures fundamental aspects of the domain architecture that are independent of marginal score fluctuations, making it suitable for both relaxed and conservative annotation settings.

Fold 1	$Q2 = 1.000$, $PPV = 1.000$
	$TPR = 0.9836$, $MCC = 0.9918$
Fold 2	$Q2 = 1.000$, $PPV = 1.000$
	$TPR = 0.9945$, $MCC = 0.9973$

Table 2: Classification metrics per fold (E-value = $1e^{-5}$)

Overall, the structural HMM displayed excellent classification properties in both folds, combining high sensitivity with perfect precision and strong resilience to dataset imbalance. Its effectiveness at correctly identifying domain-containing sequences while completely avoiding spurious hits demonstrates the strength of integrating structural alignment into profile construction. These findings support its use in large-scale annotation tasks where both accuracy and reliability are critical.

1e-5, Fold 1	MCC = 0.9918,	TPR = 0.9836
1e-5, Fold 2	MCC = 0.9973,	TPR = 0.9945
1e-9, Fold 1	MCC = 0.9918,	TPR = 0.9836
1e-9, Fold 2	MCC = 0.9973,	TPR = 0.9945

Table 3: Performance comparison at E-value thresholds 1e-5 and 1e-9.

3.2 Threshold robustness and MCC trend

To investigate the behaviour of the structural HMM across varying classification thresholds, model performance was evaluated at ten E-value cutoffs, logarithmically spaced from $1e^{-10}$ to $1e^{-1}$. The Matthews Correlation Coefficient (MCC) was selected as the primary performance metric, given its robustness in imbalanced classification settings where accuracy may be misleading.

As shown in Figure 6, MCC values remained consistently high across the entire threshold range. In Set 1, MCC stabilized at 0.989 from $1e^{-10}$ to $1e^{-7}$, then increased gradually to 0.9918 between $1e^{-6}$ and $1e^{-4}$, and further to 0.9973 at $1e^{-3}$. In Set 2, performance improved from 0.9918 to 1.000 by $1e^{-3}$ and remained perfect at $1e^{-2}$ and $1e^{-1}$.

This monotonic trend in MCC is particularly notable because it is not driven by a trade-off between sensitivity and specificity. Across all thresholds tested, the number of false positives remained consistently at zero. The gain in performance is instead attributed solely to the recovery of false negatives—true positive sequences that were initially missed under more stringent conditions. This behaviour reflects the model’s conservative nature at low E-values, where only highly confident alignments are retained, and its capacity to recover borderline positives as the threshold becomes more relaxed.

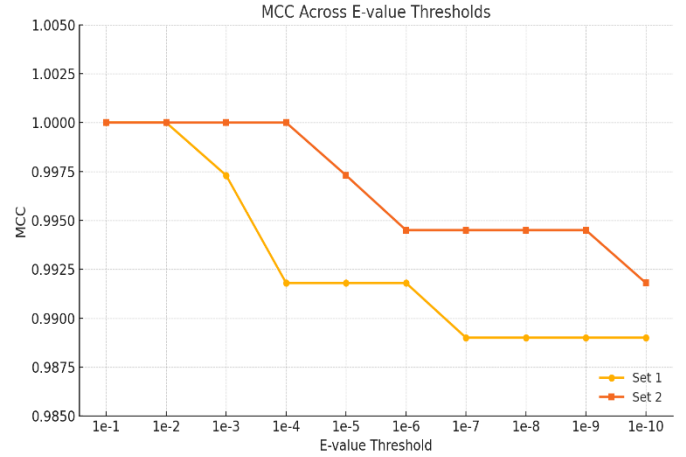


Figure 6: Matthews Correlation Coefficient (MCC) values across ten E-value thresholds for Fold 1 and Fold 2. MCC values increased monotonically as the threshold became less stringent, reaching 1.000 at $1e^{-2}$ and $1e^{-1}$ in both folds. The consistent absence of false positives across all thresholds indicates that performance gains stem from recovering additional true positives. The threshold of $1e^{-3}$ was selected as optimal, providing near-perfect classification while retaining a conservative cutoff.

Although the absolute maximum MCC values were observed at $1e^{-2}$ and $1e^{-1}$, the threshold of $1e^{-3}$ was selected as optimal because it represents the most permissive cutoff at which the model achieves near-perfect performance in both validation folds, while maintaining a conservative decision boundary. This choice provides a high-confidence compromise suitable for practical applications such as high-throughput domain annotation, where minimizing false positives is often critical.

3.3 ROC curve analysis

To provide a classification performance independently of a fixed decision threshold, a Receiver Operating Characteristic (ROC) curve was constructed by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) across all tested E-value cutoffs. The ROC curve reaches the ideal top-left corner of the plot for both validation sets, reflecting perfect sensitivity (TPR = 1) and zero false positives (FPR = 0). The corresponding Area Under the Curve (AUC) is 1.000 for both Set 1 and Set 2, indicating flawless discriminative ability across the entire threshold spectrum. This result further reinforces the robustness of the model and confirms its ability to accurately distinguish between Kunitz

and non-Kunitz sequences regardless of the classification cutoff.

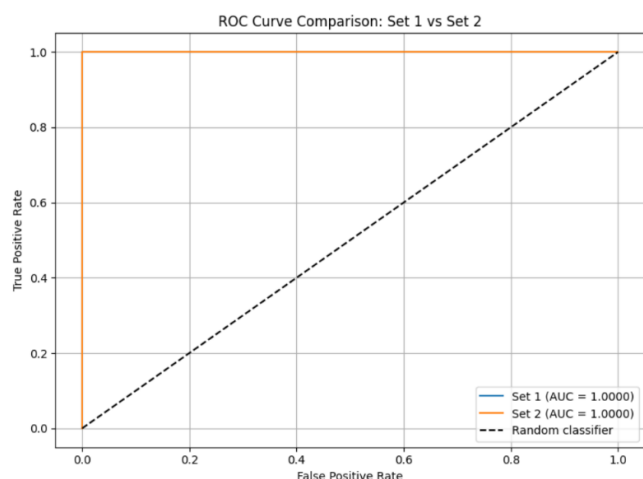


Figure 7: Comparison of ROC curves for Set 1 and Set 2. The model consistently distinguishes positive from negative sequences with perfect accuracy, as indicated by the maximum AUC values obtained in both validation folds.

3.4 Skylign and WebLogo reveal the conserved core of the domain

The conservation of amino acid positions within the structural alignment was examined through sequence logos generated with two different approaches: Skylign [12], which uses profile HMM emission probabilities to highlight conserved match states, and WebLogo [13], which depicts the raw frequency of residues across aligned sequences.

In Figure 8, the Skylign logo displays 58 match positions from the trained HMM, revealing strong conservation at several key residues. Most notably, six cysteines appear as dominant symbols at specific positions—clearly corresponding to those involved in the disulfide bridges (C1–C6, C2–C4, C3–C5) that stabilize the Kunitz fold. The extremely high information content (up to ~6 bits) and the thick cyan letters at those sites reflect both the structural relevance of these residues and the model’s ability to capture their conservation with high posterior probability. In addition to cysteines, other conserved residues such as phenylalanine (F), tyrosine (Y), and glycine (G) stand out, suggesting that these amino acids play essential roles in maintaining the domain’s compact, protease-inhibitory fold.

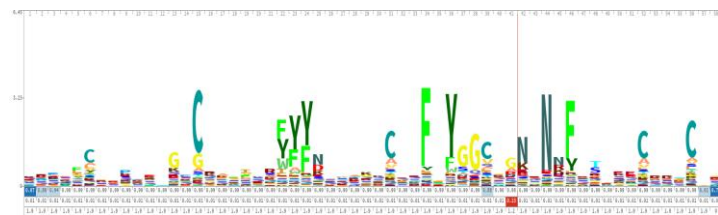


Figure 8: Sequence logo generated with Skylign from the trained profile HMM. Each column corresponds to a match state in the model, with letter height representing the information content and colour intensity reflecting the posterior probability of residue emission. The visualization emphasizes positions under strong evolutionary constraint, highlighting structurally and functionally important residues, including cysteines and aromatic amino acids.

WebLogo, shown in Figure 9, offers a complementary visualization based purely on residue frequencies observed in the alignment. This representation reinforces the conservation patterns observed in Skylign, but displays them in a more intuitive format: residues like F, Y, G, and the C-rich regions emerge as tall, clearly defined columns, especially near the domain core. The progressive narrowing of residue diversity toward certain regions indicates the presence of functionally constrained motifs.

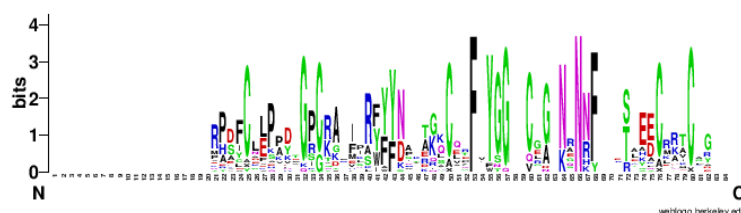


Figure 9: Frequency-based sequence logo produced from the structure-based MSA. This representation illustrates the observed amino acid distribution at each alignment position, independently of the profile model. Conserved peaks reflect recurring biochemical features across the domain, and the presence of multiple glycine, tyrosine, and cysteine residues suggests a structurally constrained and functionally specialized region.

The strong agreement between the two visualizations confirms the biological coherence of the structure-based alignment and the robustness of the trained HMM. By reflecting both sequence-level and model-inferred conservation, the logos provide compelling visual evidence that the HMM correctly captures the signature features of the Kunitz domain, which include structural disulfide bridges and aromatic or glycine-rich motifs typical of protease inhibitors.

3.5 Structural comparison of False Negative predictions

To gain a clearer understanding of model misclassifications, a structural superposition analysis was performed using UCSF ChimeraX [14]. Two sequences that were incorrectly classified as negatives—D3GGZ8 and Q8WPG5—were selected for this evaluation. Although both are annotated in UniProtKB as containing Kunitz-type protease inhibitor domains, they were not detected by the model at the optimal E-value threshold of $1e^{-3}$, which was selected based on maximum MCC performance across validation folds.

To assess whether the exclusion was due to structural incompatibility, their AlphaFold-predicted models were aligned to the canonical structure of the Kunitz domain from BPTI (PDB ID: 3TGI). The resulting superimposition, shown in Figure 10, demonstrated strong structural agreement. Specifically:

- D3GGZ8 aligned with an RMSD of 0.819 Å and alignment score of 76.9
- Q8WPG5 yielded an RMSD of 0.889 Å with a higher alignment score of 101.2

Both models preserved the hallmark $\alpha+\beta$ fold of the Kunitz domain and retained the disulfide bridge topology (C1–C6, C2–C4, C3–C5) that stabilizes the domain. Minor peripheral differences were observed, likely due to flexible termini or regions beyond the conserved core.

These findings indicate that the misclassifications were not the result of structural divergence, but rather due to borderline E-values just above the model's cutoff. This reflects the model's high specificity but also highlights the risk of excluding structurally valid domain instances when relying solely on stringent sequence-level criteria. Overall, the structural validation confirms the functional relevance of these sequences and supports the integration of 3D analysis into HMM-based domain annotation, particularly when borderline predictions occur.

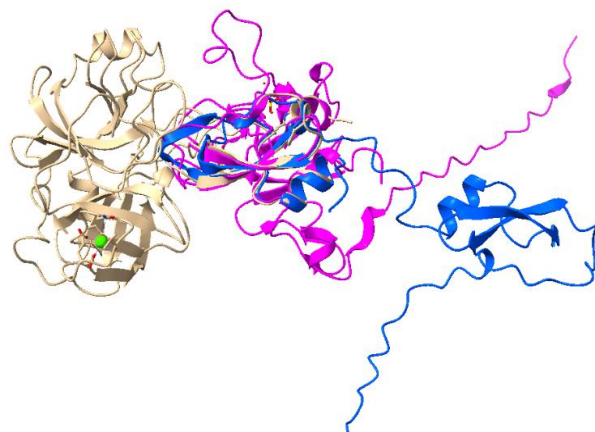


Figure 10: Structural superposition of two AlphaFold-predicted proteins misclassified as negatives by the HMM. The image illustrates the spatial correspondence between the false negative structures (D3GGZ8 in blue and Q8WPG5 in pink) and the BPTI template (PDB: 3TGI, beige). The conserved domain geometry and compact arrangement highlight a strong topological match, suggesting that subtle sequence-level variations rather than structural divergence may underlie their exclusion.

4. Conclusion

This study demonstrates the effectiveness of integrating structural alignments into the construction of profile Hidden Markov Models (HMMs) for domain detection. By training the model on structurally aligned Kunitz-type domains, we developed a classifier capable of consistently high performance, achieving near-perfect Matthews Correlation Coefficient (MCC) scores and no false positives across validation folds. The model's robustness was further validated through Receiver Operating Characteristic (ROC) analysis, which confirmed its excellent discriminative ability across a wide range of E-value thresholds. In addition, structural inspection of borderline cases showed that some misclassified sequences retained key elements of the Kunitz fold, emphasizing the importance of incorporating three-dimensional information to support domain identification beyond what purely statistical thresholds can capture. Overall, the structure-based HMM described here provides a reliable and interpretable framework for protein domain annotation. Its ability to generalize across divergent sequences while preserving biological relevance makes it a valuable approach for the detection of compact and structurally conserved domains. This methodology could be extended to other protein families where sequence divergence poses challenges to traditional annotation tools.

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